MEMS DESIGN & FABRICATION · UNIVERSITY OF SOUTH-EASTERN NORWAY · DEPT. OF MICROSYSTEMS

MEMS Piezoelectric Microphone

Redesigning a piezoelectric MEMS microphone for bioacoustic monitoring — from COMSOL multiphysics simulation and sensitivity optimization through to a full CMOS-compatible cleanroom fabrication process.

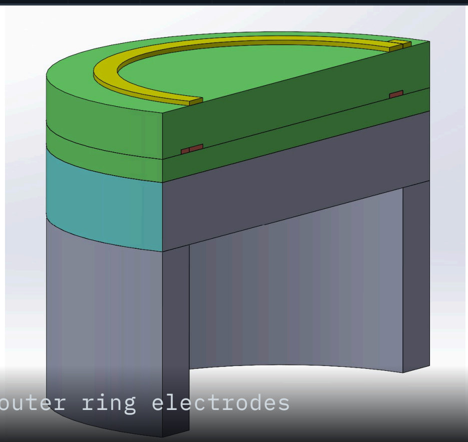

FIG.01 — Circular AlN membrane with outer ring electrodes

3x

SENSITIVITY IMPROVEMENT (1.1 → 3.21 MV/PA)

100–3500

HZ — TUNED TO ELK VOCALIZATION RANGE

10

SEQUENTIAL CLEANROOM FABRICATION STEPS

CMOS

COMPATIBLE VIA AL-GE EUTECTIC BONDING

ROLE

Team of 6. My focus: the coupled COMSOL simulation and material/geometry optimization, plus cleanroom fabrication characterization; independently authored the laser-etching section of an associated Springer book chapter.

CONTEXT

Department of Microsystems — University of South-Eastern Norway, Vestfold.

TOOLS

COMSOL Multiphysics · Photolithography · RIE / KOH Etching · Thermal Evaporation

01 WHY PIEZOELECTRIC

Conventional capacitive MEMS microphones are the industry default, but they carry real weaknesses: stiction-prone air gaps, external biasing, and parasitic capacitance from off-chip CMOS integration. **Piezoelectric transduction** sidesteps all three — no air gap, no bias voltage, linear self-powered operation — at the cost of typically lower sensitivity.

Material choice drives that trade-off. PZT offers a strong piezoelectric coefficient but poor resistivity and signal-to-noise performance. **Aluminum nitride (AlN)** is CMOS-compatible, low-loss, and highly resistive — the better engineering choice for integration, even though its lower piezoelectric coefficient caps sensitivity out of the box.

This project redesigned a published monolithic AlN-CMOS microphone design, then pushed past its baseline sensitivity to meet a specific target: reliably detecting **elk vocalizations** in the field.

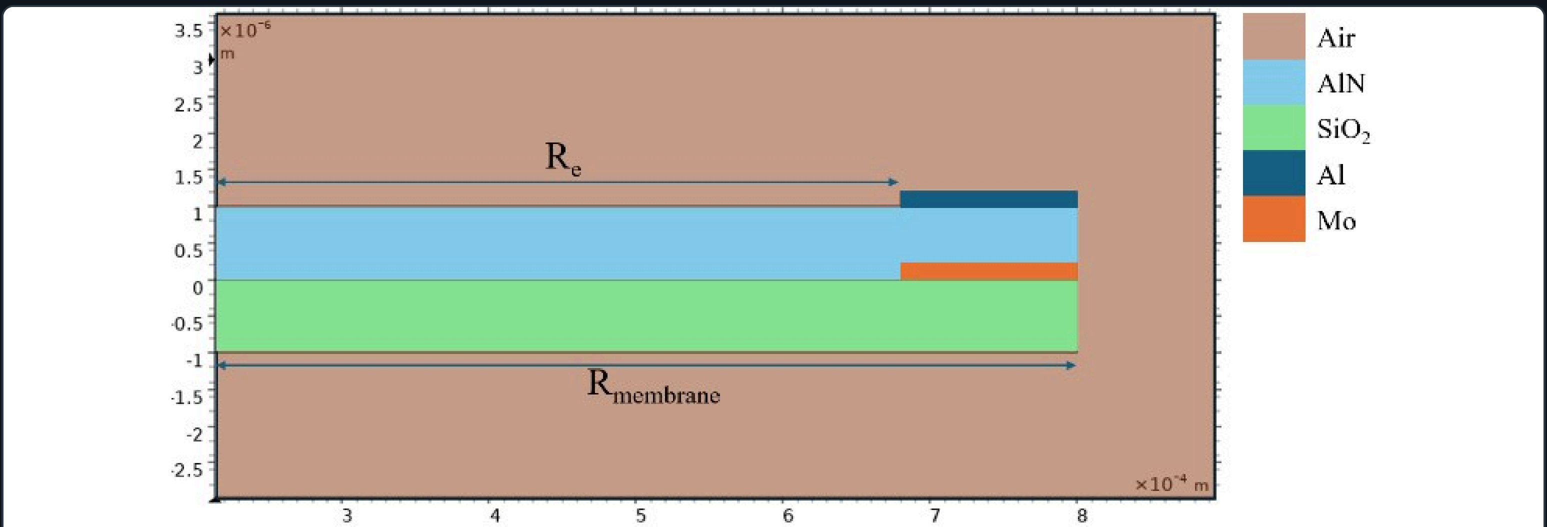

FIG.02 — Reference device geometry: circular membrane, ring electrode near the outer edge

02 SIMULATION & SENSITIVITY OPTIMIZATION

From baseline replication to a 3x sensitivity gain

The model was built in COMSOL Multiphysics as a 2D axisymmetric membrane, coupling Solid Mechanics with the piezoelectric effect to compute mechanical deformation and the resulting electric potential. The first goal was simply to replicate the reference design's behavior; the second was to redesign it for a specific application.

SENSITIVITY (OFF-RES.)

Baseline (AlN / SiO_2)

1.1 → 3.21 mV/Pa **+192%**

RESONANCE FREQUENCY

Baseline (AlN / SiO_2)

11.2 → 4.87 kHz **-57%**

Baseline (AlN / SiO_2, replicating the reference design)

Customized (ZnO / Polyimide, optimized geometry)

TARGET

Elk vocalizations concentrate in the **100–3500 Hz** band (strongest around 200–2000 Hz), and reliable detection needed a minimum sensitivity of **3 mV/Pa**. The baseline design, at 1.1 mV/Pa and an 11.2 kHz resonance well outside that band, didn't meet either requirement — which is what drove the material and geometry redesign below.

Piezoelectric material comparison

Sensitivity at 1 Pa input pressure, five candidate materials tested in the same membrane geometry.

PIEZOELECTRIC MATERIAL

ZnO

Highest sensitivity, excellent linearity (R^2=0.9999)

2.68 mV/Pa

PIEZOELECTRIC MATERIAL

$CdSO_4$

Second-highest sensitivity

2.21 mV/Pa

PIEZOELECTRIC MATERIAL

AlN (baseline)

CMOS-compatible, reference material

1.19 mV/Pa

PIEZOELECTRIC MATERIAL

PZT

High piezoelectric coefficient, but low resistivity

0.44 mV/Pa

PIEZOELECTRIC MATERIAL

$LiTaO_3$

Lowest sensitivity of materials tested

0.21 mV/Pa

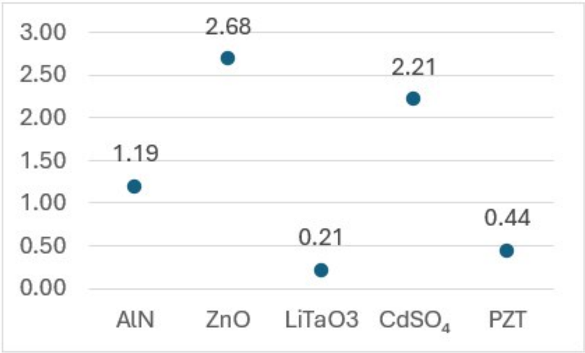

FIG.03 — Sensitivity by piezoelectric material: ZnO wins, AlN is the CMOS-friendly middle ground

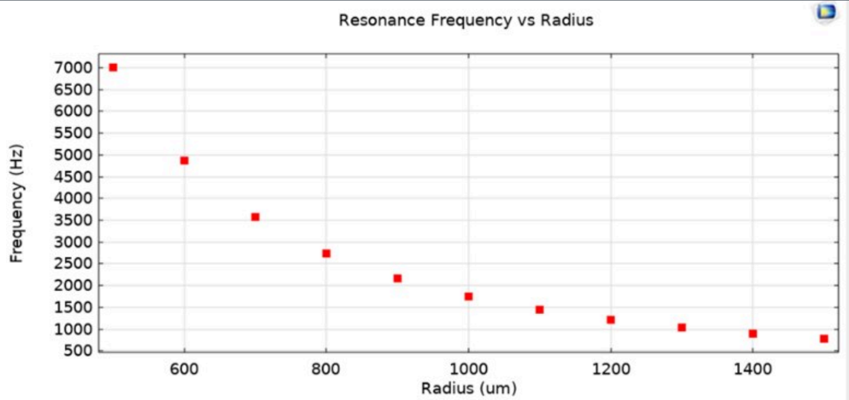

FIG.04 — Resonance vs. radius for the customized model — 600 µm chosen for a 4.87 kHz resonance

Swapping ZnO for AlN

2.4x sensitivity

ZnO delivered 2.68 mV/Pa vs. AlN's 1.19 mV/Pa in the same geometry, with excellent linearity ($R^2 = 0.9999$). Combined with a polyimide passivation layer — whose lower Young's modulus allows greater membrane deflection — this became the basis of the final design.

Geometry still mattered

Thinner SiO_2 raised sensitivity but ran into fabrication limits; AlN thickness had a clear optimum at 650 nm. Reducing electrode width and moving electrodes toward the membrane edge also improved sensitivity — at the cost of a lower resonance frequency and tighter fabrication tolerances.

03 FABRICATION PROCESS

A CMOS-compatible process, front to back

Simulation results only matter if the design can actually be built. The fabrication process was split into a front-end (all MEMS structuring on the sensor wafer) and a back-end (eutectic bonding to a CMOS carrier wafer), sequencing ten process modules from bare silicon to a released, CMOS-ready diaphragm.

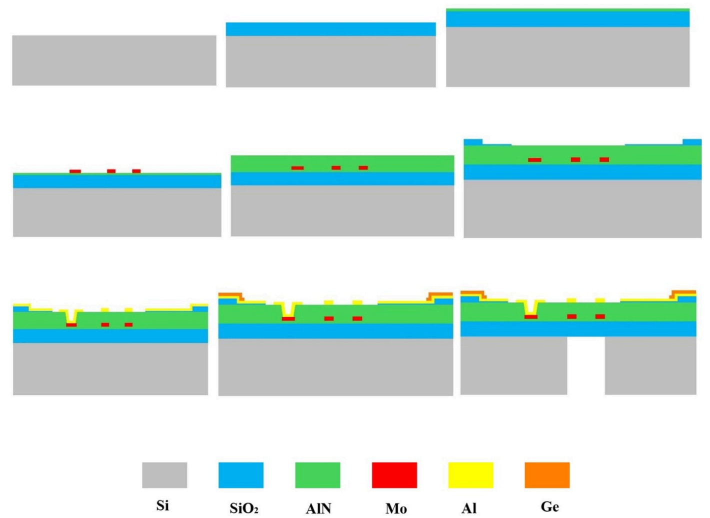

FIG.05 — Full fabrication sequence, bare silicon to released acoustic port

01

RCA clean & oxidation

Standard RCA clean, then wet thermal oxidation grows a ~400nm SiO_2 base layer.

02

AlN seed + Mo electrode

Reactive-sputtered AlN seed layer, then Mo bottom electrode patterned by RIE.

03

Main AlN layer

~1μm c-axis-oriented AlN sputtered as the active piezoelectric transduction layer.

04

SiO_2 standoffs

Two-step PECVD + lift-off builds dual-height standoffs for the bonding cavity.

05

Via + top electrode

ICP-RIE opens a via to the buried Mo electrode; Al top ring electrode patterned by etch-back.

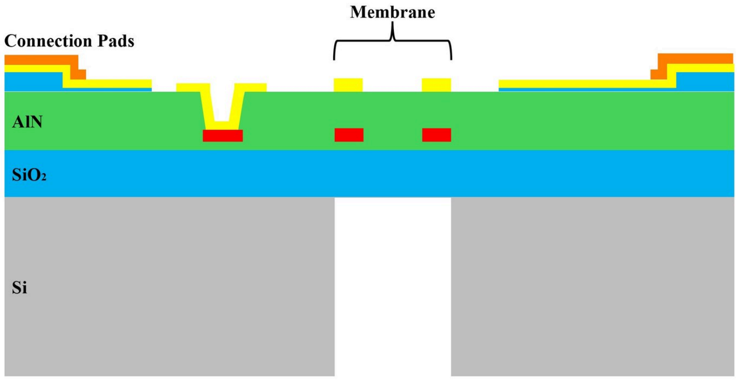

FIG.06 — Final layer stack: Si → SiO_2 → AlN → electrodes → Ge bonding pads

Back-end: getting to CMOS

Al/Ti/Ge bonding pads enable **Al–Ge eutectic bonding** to a CMOS carrier wafer — titanium bridges the poor native adhesion between Al and Ge. A final backside **DRIE etch** cuts completely through the silicon to release the diaphragm and open the acoustic port beneath it, completing a process that's scalable and CMOS-compatible end to end.

HANDS-ON VALIDATION

The individual process steps in this flow — photolithography, dry (RIE) and wet (KOH) etching, and thermal-evaporation metal deposition — were separately practiced and characterized on test silicon wafers. Measured KOH etch rates matched the literature reference of ~46 μm/h at 70°C within a few percent, and profilometry/ellipsometry measurements confirmed consistent, controlled layer thicknesses across every step — the process control this microphone's fabrication flow depends on.

04 PUBLICATION

Contributing author, published book chapter

Sp

Green Etching Techniques for MEMS Applications

Photoresist Technology in Microsystems: Principles, Processes and Applications

Contributed the section on **laser-based etching for MEMS fabrication** — comparing excimer, femtosecond, and CO_2 laser systems for precision micromachining, alongside the hybrid and AI-driven dry-etching techniques covered in the wider chapter.

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05 ENGINEERING DECISION

Trading CMOS-standard material for a 2.4× sensitivity gain

AlN is the conventional choice for CMOS-integrated piezoelectric MEMS — it's what the reference design used. Hitting the elk-vocalization sensitivity target meant deciding whether that convention was worth keeping.

CONSIDERED

AlN (the baseline/reference material) — CMOS-compatible and the industry-standard choice, but capped sensitivity at 1.19 mV/Pa in the tested geometry, short of the 3 mV/Pa target.

CHOSEN APPROACH

ZnO piezoelectric layer with a polyimide passivation layer, replacing the AlN/SiO_2 baseline stack.

WHY

ZnO delivered 2.68 mV/Pa vs. AlN's 1.19 mV/Pa in the identical geometry — a 2.4× gain, with excellent linearity ($R^2 = 0.9999$) across the five materials tested. Polyimide's lower Young's modulus allows greater membrane deflection than the baseline SiO_2 passivation, adding further sensitivity on top of the material swap. The trade-off is giving up AlN's more standard CMOS-integration track record for the performance needed to hit the target band.

06 CHALLENGES

Where simulation gains met fabrication reality

Sensitivity improvements that ran into fabrication limits

Thinner SiO_2 raised simulated sensitivity, but only up to the point where it became impractical to fabricate reliably. Reducing electrode width and moving electrodes toward the membrane edge also helped sensitivity — at the cost of a lower resonance frequency and tighter fabrication tolerances. Every geometry change in the simulation had to be checked against what the ten-step cleanroom process could actually hold, not just what COMSOL said was optimal.

The redesigned material stack wasn't part of the characterized fabrication flow

The process modules that were actually built and measured on test wafers — photolithography, RIE/KOH etching, thermal evaporation — validate the AlN-based CMOS-compatible flow. The ZnO/polyimide stack that wins on simulated sensitivity introduces different deposition and etch chemistry that wasn't part of that characterization, which matters for anyone treating the 2.4× gain as fabrication-ready rather than simulation-validated.

07 REFLECTION

What I'd do with another pass at this

NEXT STEPS

The biggest gap between this project's simulation and fabrication halves is exactly the point above: the ten process modules characterized on real wafers validate the AlN-compatible flow, not the ZnO/polyimide stack that the sensitivity optimization actually recommends. A natural next step would be characterizing ZnO deposition and polyimide patterning on test wafers the same way AlN etch rates were characterized against the ~46 μm/h literature reference, to find out whether the simulated 2.4× gain survives contact with the actual process.

08 SUMMARY

What this project demonstrates

- Multiphysics FEA: coupled solid-mechanics and piezoelectric simulation in COMSOL
- Materials and geometry trade-off analysis for a specific performance target
- Hands-on cleanroom process skills: photolithography, RIE/KOH etching, thermal evaporation
- Validating simulation against a published reference design before redesigning
- Translating a simulated design into a real, CMOS-compatible fabrication flow
- Technical writing at publication standard, including a contribution to a Springer book chapter

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